

Teng Zhang

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Professional Summary

Self-motivated geospatial data scientist with a focus on interdisciplinary applications, dedicated to advancing spatial awareness and critical thinking in both research and teaching.

- Proficient in large-scale data analysis and management using R, Python and GIS, with extensive experience in automated data-crawling and machine learning.
- Extensive teaching experience in both in-person and online settings, recognized as 'Excellent' online geography instructor at MSU
- Committed to integrating AI-driven approaches to enhance research, teaching, and public understanding of complex geo-social science challenges.

Education

2023 Michigan State University, East Lansing, MI

PhD in Geography (Spatial analysis of human rights records)

Dissertation: *Unsafe Space: Spatial Analysis of Indonesian Genocide in Java*

Integrating digitalized local knowledge with large-scale post-WWII maps, this dissertation applies innovative spatial analysis tools, including Bayesian spatial regression, hierarchical regression (multilevel models), and spatial simulation, to the quantitative study of human rights records.

Dissertation Advisors: Prof. Ashton Shortridge and Prof. Siddharth Chandra

2013 Peking University, Beijing, China

Master of Science in Geography (Urban and Regional Planning)

Thesis: Environment Effect of the Industrial Geography Transformation in China

2010 Renmin University of China, Beijing, China

Bachelor in Public Administration (Environmental Economics and Management)

Selected Research and Professional Experience

2024 –

Postdoc Research Fellow

Dedman Law School

Southern Methodist University, Dallas, Texas

- Collaborated with attorneys and professors at law school to identify rural areas across various US states with insufficient legal resources, and designed a metric system based on equilibrium models to evaluate the severity of legal deserts
- Developed automated web-crawler to collect and process over 300,000 court caseload records across multiple states, and created R pipelines to clean, geocode, and integrate the records with demographic data
- Applied network analysis to map lawyer–court interaction pairs, identified clusters of

legal resources, and traced their shifts over decades to highlight structural access-to-justice issues at the state level

2016 – 2023 **Research Assistant**

Dept. of Geography, Environment and Spatial Science
Michigan State University, East Lansing, Michigan

- Performed R/Python-based spatial and network analysis of socio-economic data from Indonesia, China and Colombia, converted the results to audience friendly report with innovative data visualization techniques
- Applied Bayesian modeling and spatial statistics to examine patterns of mass violence and regional inequality

2014 – 2016 **Senior Analyst** in Office/Warehouse Market Research Sector

Jones Lang LaSalle, Guangzhou, China

Work in Process and Peer-reviewed Articles

In Prep/Submitted (selected):

Zhang, T., Chandra, S., Untangling Mass Violence in Indonesia: A Geographic Perspective on the 1965-1966 Killings. (Accepted by *Bijdragen tot de Taal-, Land- en Volkenkunde*)

Zhang, T., Chandra, S., An Act of Killing? Re-examining the Roles of the Persecutors in the Indonesian Communist Purge with Evidence from Disaggregated Geographic Analysis (submitted to *Political Geography*)

Zhang, T., Chandra, S., Should Faction Leaders in the Military Be Held Accountable for Mass Violence? A Multilevel Linear Model Analysis of the Indonesian Mass Killings in 1965-66 (*in preparation*)

Zhang, T., Davis, A., Measuring Legal Deserts: A Panel Data Equilibrium Model Approach (*in preparation*)

Published:

Chandra, S., Zhang, T., 2023, Glimpses of Indonesia's 1965 Massacre through the Lens of the Census: the Role of Civilian Organizations in the Mass Anti-Communist Killings of 1965-66 in East Java, Indonesia. *Indonesia*.

Zhang, M., Zhang, T., Xiao, Z. and Chai, Y., 2021. Property rights redistribution and the spatial evolution of the Chinese danwei compound: a case study in Beijing. *Journal of Housing and the Built Environment*, pp.1-18.

He, Canfei., Zhang, T. and Wang, Rui., 2012. Air quality in urban China. *Eurasian Geography and Economics*, 53(6), pp.750-771.

Liu, Che., Zheng, Chengyang., Zhang, T., Relation of Bird Species Geography and Environmental Factors, *Acta Universitatis Pekinensis*, 2014, 3 (in Chinese: 中国鸟类物种丰富度的地理格局及其与环境因子的关系, *北京大学学报*, 2014, 3)

He, Canfei., Zhou, Yi., Zhang, T., Industrial Transfer and its Environmental Effects in China, *Study of Urban and Environment*, 2014, 1(1), 41-55 (in Chinese: 中国产业转移及其环境效应, *城市与环境研究*, 2014, 1(1), 41-55)

He, Canfei., Zhang, T., Environmental Regulation and its Effect on Air Pollution in China, *Journal of Natural Resources*, 2013, 28(10), 1651-1663 (in Chines: 环境规制效果与城市空气质量, *自然资源学报*, 2013, 28 (10),1651-1663)

Ma, J., Zhang, T., Chen, Z.Y., Immigration for Environment Protection: Case in Dongxiang Gansu , *Journal of Gansu Normal Colleges*, 2010, 2 (生态脆弱地区移民情况及问题浅析 • 以甘肃省临夏回族自治州东乡族自治县为例, *甘肃高师学报*, 2010, 10)

Selected Conferences and Invited Talks

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| Mar. 2025 | "Network Analysis of Legal Deserts: Interactions Between Indigent Defendants and Lawyers", AAG 2025 Annual Meeting, Detroit, MI |
| Nov. 2024 | "Where's My Lawyer? A Spatial Analysis of Criminal Attorneys in Texas", 2024 American Society of Criminology Annual Meeting, San Francisco, CA |
| Oct. 2024 | "Legal Resource in Texas and its Shortage", 2024 Southwest Regional Division Meeting, San Marco, TX |
| Apr. 2024 | "Martyrs or Mercenaries? — Motivation of the Military in the Mass Murders, a Multilevel-linear Model", AAG 2024 Annual Meeting, Honolulu, HI |
| Apr. 2023 | "Community's Cohesion and Resistance in the Indonesian Anti-Communist (Anti- PKI) Purge of 1965-1966: Variables", AAG 2023 Annual Meeting, Denver, CO |
| Jun. 2022 | "A spatial analysis of the local community in East Java in Indonesian Genocide of 1965-1966", <i>AIFIS-MSU 2022 Conference on Indonesian Studies</i> , East Lansing, MI |
| Feb. 2022 | "The Role of Military Units in the Mass Killings of 1965-66 in East Java, Indonesia: Spatial Analysis" (Co-author), <i>AAG 2022 Annual Meeting</i> , virtual |
| Jun. 2021 | "Willing to Kill or Ordered to Kill: Re-examining the Roles of Persecutors with Evidence From Disaggregated Geographic Analysis", <i>AIFIS-MSU 2021 Conference on Indonesian Studies</i> , East Lansing, MI |
| Oct. 2020 | • "Military and Civilian Involvement in the Killings of 1965-66 in East Java: A Spatial Approach", <i>69th Midwest Conference on Asian Affairs (MCAA 2020)</i> , East Lansing, MI |
| Apr. 2019 | "The Very Inexpensively Built 'Big Beautiful' Wall inside Chinese Post-socialist Neighborhoods", <i>AAG 2019 Annual Meeting</i> , Washington, D.C. |
| Apr. 2018 | "Moving slums in China: Documentary and Discussion", AAG 2018 Annual Meeting, New Orleans, LA |
| Oct. 2017 | "Geography of Cuisines in Beijing----- description and potential risks", <i>ESPP Fall Research Symposium 2017</i> , East Lansing, MI |
| Nov.2015 | "Spatial Evolution in Huagong Da'yuan Danwei Compound", <i>The 3rd Int. Conf. on Social Policy & Governance Innovation</i> , Guangzhou, China |
| Jan. 2013 | "Air Pollution and Implementation of Environmental Regulation in Chinese Cities (Poster)", <i>Youth Forum of China Environment Protection 40 Years Forum</i> , Hong Kong |
| Dec. 2012 | "A Chronicles of Stoves in Rural Gansu", <i>Shell Co. Graduates Rural Area Stove Symposium</i> , Beijing, China |
| Jun. 2012 | "Distance and Growth of Urban GDP in North China (Poster)", <i>7th Sino-Korea-Japan Joint Conference on Geography</i> , Changchun, China |

Selected Teaching Experience

Undergrad Courses	Instructor of Intermediate GIS • Cartography, geodesy basics, Raster/Vector data models and analysis Instructor of World Regional Geography (online) and Physical Geography (online), Geography of the US and Canada (online) Teaching Assistant of Problems in Geographic Information Science, People and Environment, Spatial Analysis, etc.
Grad Courses	Teaching Assistant of Spatial Data Analysis, Geospatial Technology
Online Certificate	Instructor of Introduction of Cartography, Instructor of Criminal Justice Applications in GIS, Instructor of Geographic Information System

Skills and Hobbies

Technical	ESRI ArcGIS/QGIS, R & Python based (Spatial) data analysis and science: Stan, PyTorch, Geo-simulation (Agent-based Model), Google Earth Engine, Quantitative data analysis tools: multilevel regression, Bayesian regression, network analysis, Econometrics, UCINET, PostgreSQL, Computational Modeling Web page crawler (Multi-threaded)
Skills	Photoshop and Mandarin (native tongue)
Interests	Financial Market, Budget travelling, Sailing, Erhu, and Taekwondo

Grants, Fellowships and Awards

2024	Spatial and Demographic Aspects of the Killings of 1965-66 in Indonesia: Methods and Implications for Mass Atrocity Prevention, <i>Promoting Peacemaking and Peacebuilding in Communities Experiencing Conflict Mini Grant</i> , Michigan State University, Co-Investigator
2023	Michigan State University Dissertation Completion Fellowship
2022	Kussy Scholarship for Study of Genocide
2022	S.C. Lee Best Paper Competitions – 3 rd Place
2019	Dr. Delia Koo Global Student Scholarship
2019	AAG Annual Meeting Poster Competition Award (Eurasian Environments poster session)
2017	MSU ESPP Research Symposium Network Fellowship
2017	NCUSCR Travel Fund for US Foreign Policy Colloquium
2012	UNEP-ISED Beijing Green Future Environment Foundation Lyumiao Fellowship
2012	Hong Kong Jockey Club Travel Fund for Youth Forum of China Environment Protection
2012	Shell Co. Graduates Rural Energy Saving Investigation Scholarship
2007	National Undergraduate Innovative Experimental Program (China) -RUC Fellowship

References

Prof. Ashton Shortridge (ashton@msu.edu)	Prof. Siddharth Chandra (chandr45@msu.edu)
Dr. Andrew L. B. Davies (albdavies@smu.edu)	Prof. Kyle Evered (ktevered@msu.edu)